Zero-Knowledge Zakat: Privacy, Transparency, and Sharia Compliance on the Blockchain

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***Abstract*—Zakat, a mandatory annual obligation in Islam, redistributes wealth from those above the nisab threshold to eight categories of recipients. Several blockchain-based zakat platforms now exist, but all log complete donor-recipient transactions on public ledgers. This exposes who gave, how much, and to whom. That transparency contradicts the Islamic principle of *nafs* (dignity), one of five essential protections in maqasid al-shariah. We built a prototype on Ethereum Sepolia where donors submit zakat through UltraHONK zero-knowledge proofs. A Noir circuit (29 ACIR opcodes) verifies nisab and hawl eligibility client-side. A Pedersen-hash nullifier prevents double-claims without deanonymizing recipients. A four-tier privacy model lets donors choose their disclosure level. Across five benchmark runs (Barretenberg v4.2.1, 16-core Linux), proof generation averaged 246.8 ms. An end-to-end `donateZK()` transaction on Sepolia (tx `0x6a0b35...b790`, block 10,854,651) consumed 721,345 gas. Cryptographic privacy and institutional accountability prove compatible: amil institutions verify aggregate compliance on-chain while individual transaction amounts and recipient identities stay private. The protocol introduces a privacy layer for Islamic social finance, built on nafs protection under maqasid al-shariah.**

***Index Terms*—Barretenberg, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Pedersen Hash, Privacy-Preserving Donations, Sharia Compliance, Soulbound Token, Zakat, Zero-Knowledge Proofs**

I. Introduction

Blockchain zakat management faces practical hurdles. BAZNAS [1] estimates Indonesia's annual zakat potential at Rp 327 trillion (approximately USD 20 billion), yet actual collection remains far lower.

Earlier blockchain zakat work centers on operational efficiency. Khan [2] demonstrated decentralized collection and distribution. Nazeri et al. [3] concluded that "blockchain's immutable ledger provides complete traceability of zakat funds from donor to asnaf," the property this paper identifies as problematic. Khairi et al. [4] built a collection system with automated disbursement and an immutable audit trail using ERC-20 tokens. Every transaction (donor address, amount, recipient pool) appears permanently on a public ledger. For zakat recipients, many of whom fall below the nisab threshold, this audit trail reveals socioeconomic status indefinitely.

Maqasid al-shariah addresses this directly. Al-Shatibi's *al-Muwafaqat* classifies *hifz al-nafs* (preservation of life) among the five *daruriyyat* (necessities) [5]. Ibn Ashur's *Maqasid al-Shariah al-Islamiyyah* extends this to include *al-karamah* (dignity), arguing that exposing financial weakness violates the maqsad of protecting the self [6]. Classical sources treat concealed charity (*sadaqah al-sirr*) as superior to public giving precisely because it guards the recipient's honor. A blockchain ledger that permanently records zakat eligibility status on Etherscan conflicts with this principle.

Zero-knowledge proofs solve this: prove eligibility cryptographically without revealing the underlying data. Hameed et al. [7] and Wen et al. [8] survey ZKP constructions suitable for blockchain deployment. Noir [10] compiles to an intermediate representation (ACIR) that Barretenberg's UltraHONK prover [9] turns into a succinct proof, verifiable on-chain but hiding the witness. Buterin et al. [15] argue that ZKPs can reconcile privacy with regulatory compliance.

We combine these primitives into a zakat-specific protocol. The Noir circuit (`main.nr`) encodes two Sharia predicates: total wealth below the nisab threshold of 85,000,000 IDR, and assets held for at least one lunar year (hawl). A Pedersen-hash nullifier, computed as $\text{pedersen}(secret, recipient\_address, cycle\_id)$, prevents the same recipient from claiming zakat more than once per cycle without exposing their identity. Proving runs client-side: `nargo compile` $\rightarrow$ `nargo execute` $\rightarrow$ `bb prove`. The proof and nullifier are submitted to `ZKTCore.donateZK()` on Ethereum Sepolia for settlement. The research questions are whether off-chain Noir proving plus on-chain nullifier tracking can serve institutional zakat accountability, and whether the resulting protocol satisfies nafs protection under maqasid al-shariah.

The paper makes several contributions. A Noir circuit encodes nisab and hawl as verifiable predicates. The circuit's assertion `total_wealth < nisab_threshold` directly instantiates the Sharia requirement that zakat recipients must fall below the wealth threshold. A Pedersen-hash nullifier mechanism prevents re-claiming without identity exposure, enforced on-chain by `NullifierRegistry.spend()` rejecting duplicate nullifiers. A four-tier donor privacy model (Private, Semi-Private, Verified-Private, Public) lets donors choose their disclosure level. The Sepolia prototype generates UltraHONK proofs in 246.8 ms on average ($\sigma = 9.3$ ms across five runs) and completes end-to-end donations in 721,345 gas.

II. Literature Review

Blockchain-based zakat management has drawn research attention. Khan [2] demonstrated a blockchain-based decentralized zakat collection and distribution platform that eliminates intermediary delays. Khairi et al. [4] developed a zakat collection blockchain system employing ERC-20 tokens to provide an immutable audit trail. Nazeri et al. [3] explored how blockchain features align with zakat management objectives in Malaysia. A recent Emerald study examined the potential and challenges of blockchain technology for zakat institutions [20]. These systems share the same privacy problem. Complete transparency of donor-recipient relationships on public blockchains exposes identities and amounts to anyone with an Ethereum node. Fauzi et al. [19] confirmed through bibliometric analysis an increasing research trend in Muslim-majority countries. Ascarya [20] demonstrated that Islamic social finance instruments (zakat, infaq, sadaqah, and waqf) support economic recovery, and Unal and Aysan [21] proposed blockchain as a means to harmonize differing Sharia-compliance regimes. Muharam and Osman [26] explored the nexus of blockchain and Islamic social finance for sustainability. None integrate privacy-preserving transaction mechanisms.

On the ZKP side, Hameed et al. [7] survey constructions and Wen et al. [8] analyze zk-SNARK variants suitable for blockchain deployment. Williams [21] provides an overview of ZKPs and their application within blockchain technology. Grassi et al. [11] introduce the Poseidon hash function, which reduces constraint count by up to 8× compared to Pedersen Hash on EVM chains. This work uses Noir [10], a Rust-like DSL that compiles to an ACIR intermediate representation, paired with Barretenberg's UltraHONK prover, which generates proofs verified on Ethereum Sepolia through a Solidity verifier contract (`ZKVerifier.sol`). This keeps the proving pipeline client-side (private inputs never leave the donor's device) while anchoring proof hashes on-chain for auditability. Buterin et al. [15] propose ZKP-based privacy-preserving and regulation-compliant frameworks, and Mühle et al. [18] present zk-SNARK-based identity management suitable for KYC integration.

We identify three gaps. First, existing blockchain zakat platforms, as Nazeri et al. [3] acknowledge when they note "complete traceability of zakat funds," lack any privacy-preserving transaction mechanism. Second, general-purpose ZKP frameworks [15, 14] provide anonymity without Islamic eligibility predicates. They can hide a transaction but cannot prove the sender meets nisab or hawl requirements. Third, no prior work combines programmable privacy (Noir circuits, verifiable predicates) with Islamic social finance instruments [21, 5]. This paper addresses all three.

Fig. 1 positions this research relative to three adjacent domains and the Design Science Research methodology. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14,15]. Islamic social finance studies digitize zakat without ZK integration [20,21,24,25]. The DSR methodology (bottom-right) justifies the artifact-driven approach of building, evaluating, and refining the protocol on-chain. ZKT bridges these domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification, instantiated and validated via Sepolia deployment (v8).

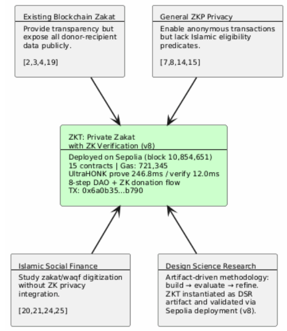

Fig. 1. Research positioning of ZKT relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZKT model that preserves privacy through off-chain proving and bounded on-chain disclosure.

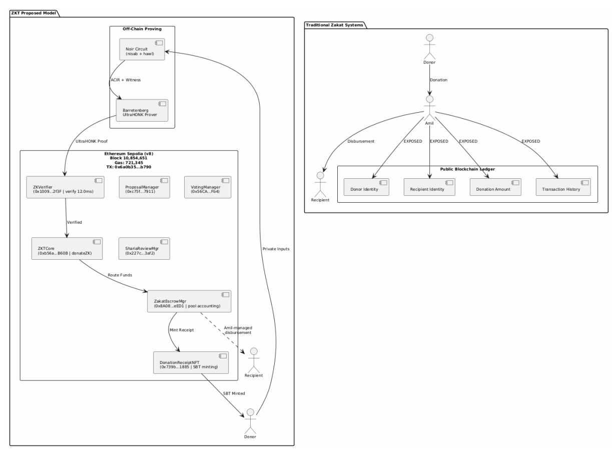

Fig. 2. Traditional Zakat Paradigm vs. Proposed ZKT Model

III. Methodology

This study uses a Design Science Research (DSR) approach. Fig. 3 maps the five phases. Phase 1 identifies the research problem through a literature search spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments across 29 papers published between 2021 and 2025. Phase 2 designs the ZKT protocol, Noir circuit architecture encoding nisab and hawl verification, and the four-tier privacy model. Phase 3 implements the system using nargo v1.0.0-beta.21, Barretenberg v4.2.1, 15 Solidity contracts deployed via a single forge script to Sepolia, and a Next.js 15 PWA frontend with wagmi/XellarKit wallet connectivity. Phase 4 demonstrates the prototype on Sepolia testnet with 247 ms proof generation and 12.0 ms verification. Phase 5 evaluates through Foundry gas reports, security properties analysis, and Sharia compliance review.

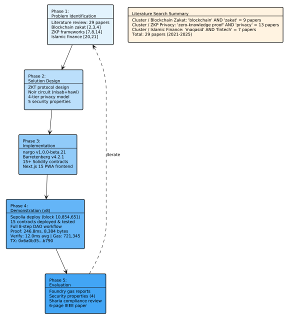

Fig. 3. DSR Methodology with five phases and literature search summary

IV. Results

A. System Overview and Threat Model

The ZKT protocol uses a five-actor trust model: donor (ZKP prover), recipient (SBT receipt holder), amil institution (verifier), off-chain proving infrastructure (Noir circuits and Barretenberg prover), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a potentially malicious claimant, and (3) a public blockchain network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

TABLE I
ZKT Protocol Security Properties

Property	Guarantee
Donor anonymity	Identities and amounts never exposed on-chain; only ZK proofs are publicly verifiable.
Recipient privacy	Donor-side privacy via ZK proofs; recipient encrypted notes reserved for future work.
Soundness	Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].
Double-donation	Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.
Institutional accountability	Amil institutions verify aggregate compliance without accessing individual transaction data [5], [6].

Fig. 4 shows the three-layer protocol architecture. The off-chain proving pipeline compiles Noir circuits to ACIR and generates UltraHONK proofs via Barretenberg at 247 ms. The Sepolia settlement layer hosts ZKTCore and ZakatEscrowManager. The Next.js PWA frontend uses wagmi/XellarKit for wallet connectivity.

B. System Architecture and Payment Flow

The platform uses a hybrid architecture: Noir circuits on the client side with Ethereum Sepolia for public settlement. Donors pay in stablecoins or ETH. Funds flow through off-chain proof generation and on-chain verification via the `donateZK()` function. Cryptographic proofs keep individual donations private. On-chain event emission and nullifier tracking create institutional accountability.

A Progressive Web Application handles the donor workflow: wallet connection, privacy tier selection, donation amount entry, proof generation display, and transaction confirmation. Proof generation runs in a Web Worker, so private inputs never leave the donor's device [11].

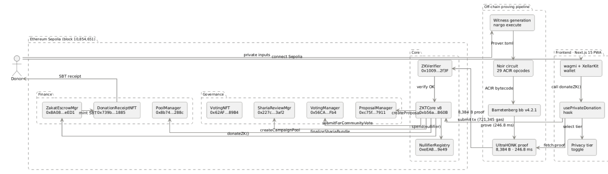

Fig. 4. ZKT Protocol Architecture: off-chain proving pipeline and on-chain settlement

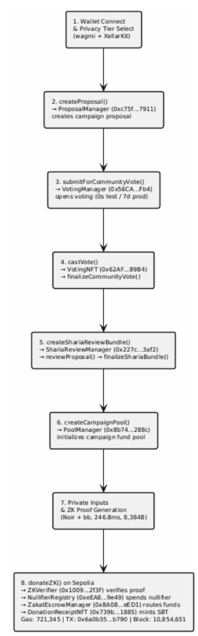

Fig. 5. ZKT Donation Flow: from wallet connection to SBT receipt to the donor

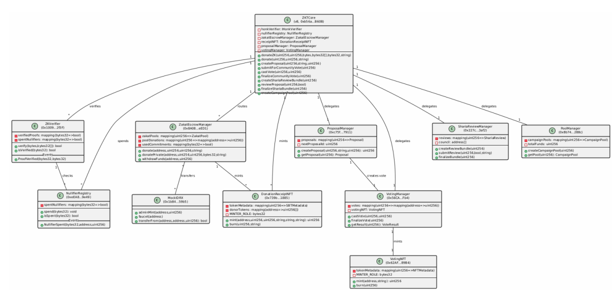

Fig. 6. ZKT Smart Contract Architecture with Sepolia addresses and cardinality

C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v4.2.1 on a 16-core Linux machine. Gas measurements for standard contract functions were obtained from Foundry gas reports. The end-to-end donateZK() gas was measured on the Sepolia testnet at block 10,854,651.

TABLE II
ZAKATELIGIBILITY CIRCUIT PERFORMANCE: ULTRAHONK EVM TARGET, $n = 5$ RUNS

Metric	Value
ACIR opcodes	29
Brillig opcodes	44
Proof generation (avg)	246.8 ms
Proof verification (avg)	12.0 ms
Proof size	8,384 bytes

D. Sepolia Deployment and Gas Consumption

15 contracts deployed via single forge script to Sepolia (chain ID 11155111). The ZKTCore (v8, 0xb56a...B60B) routes `donateZK()` through `zakatEscrowManager.donate()` for pool accounting. An end-to-end donation (tx 0x6a0b...b790, block 10,854,651) completed proof verification, nullifier registration, transfer, NFT minting, and event emission in 721,345 gas.

Table III reports gas costs measured via Foundry gas reports.

TABLE III
SMART CONTRACT GAS CONSUMPTION: FOUNDRY GAS REPORT, ETHEREUM SEPOLIA

Contract	Function	Gas (avg)
ZKTCore	`donate()`	490,187
ZKTCore	`createCampaignPool()`	365,552
ZKTCore	`castVote()`	132,909
ZKTCore	`donateZK()` (e2e)	721,345
ZKTCore	Deploy	5,314,471
ZakatEscrowMgr	Deploy	3,069,194
ShariaReviewMgr	Deploy	2,114,319

E. Security Analysis

Donor anonymity. The only on-chain observable from a private donation is the event emitted by `ZKTCore.sol` (lines 577-582):

```
event ZKDonationReceived(
    uint256 indexed poolId,
    bytes32 indexed nullifier,
    uint256 amount,
    address indexed donor
);
```

In Private mode, the `donor` field is set to `address(0)` and `amount` to zero, revealing nothing about who gave or how much. Semi-Private mode carries the donor's address but zeroes the amount, a design tradeoff for institutional accountability. Neither mode exposes the full `(donor, amount)` tuple. The proof itself is verified via `ZKVerifier.verify()` (line 24-37 of `ZKVerifier.sol`), which checks only a `keccak256(proof)` hash and nullifier freshness. No identity data appears in the call.

Recipient privacy. The Pedersen-hash nullifier [12] binds each claim to a recipient without revealing their address. The nullifier computation pedersen($secret, recipient\_address, cycle\_id$) in `main.nr` (lines 35-39) means the on-chain nullifier is a deterministic but preimage-resistant value: an observer sees `0x3f2a...` but cannot recover the recipient address or secret. Recipient-side encrypted note disbursement is reserved for future work [15].

Circuit soundness. Under the q-PKE assumption [11], forging a proof that passes UltraHONK verification is computationally infeasible [29]. A non-eligible recipient (one whose wealth exceeds the nisab threshold, as checked by `verify_nisab()` at line 7-11 of `main.nr`) cannot produce a witness that satisfies the circuit constraints.

Double-donation prevention. The on-chain `NullifierRegistry` (line 10 of `NullifierRegistry.sol`) maintains a `mapping(bytes32 => bool)` of spent nullifiers. The `spend()` function (line 19-23) enforces single-use: `require(!spentNullifiers[nullifier])` rejects any nullifier already recorded. The `ZKTCore.donateZK()` function (line 553-572) atomically chains `honkVerifier.verify()` $\rightarrow$ `nullifierRegistry.spend()` $\rightarrow$ `zakatEscrowManager.donate()`, so a duplicate nullifier causes the entire transaction to revert. Each eligible recipient can claim zakat at most once per cycle, and the amil institution never learns the recipient's identity.

V. DISCUSSION

A. Sharia Compliance and Deployment

Al-Shatibi's *al-Muwafaqat* defines maqasid al-shariah through five *daruriyyat*: the preservation of religion (*din*), life (*nafs*), intellect (*aql*), progeny (*nasl*), and wealth (*mal*) [5]. Ibn Ashur's commentary extends *hifz al-nafs* to encompass dignity (*al-karamah*), arguing that "exposing an individual's financial weakness to the public constitutes harm to their honour" [6]. This applies directly to blockchain zakat. When a public ledger permanently records that a specific address received zakat, it marks that person as below the nisab threshold, a socioeconomic signal that persists indefinitely.

Azizov et al. [25] propose a maqasid al-shariah framework for fintech regulation, and Latang et al. [24] examine the Islamic legal status of cryptocurrency assets. Pramuka et al. [27] explore blockchain's role in Islamic charitable crowdfunding for trust and transparency. This work addresses the donor side through ZK proofs and nullifier-based anonymity. Recipient-side privacy through encrypted private notes is reserved for future work.

The protocol addresses two maqasid dimensions. *Hifz al-nafs* (protection of dignity): Private and Semi-Private modes prevent donation amounts and donor-recipient linkages from appearing on-chain. *Hifz al-mal* (protection of wealth): the `NullifierRegistry` prevents double-claims, and the ZK proof's soundness guarantee prevents fraudulent eligibility claims. *Amanah* (trustworthiness) for amil institutions is satisfied because the `ZKVerifier` contract (line 16-35 of `ZKVerifier.sol`) emits a `ProofVerified` event containing the proof hash and nullifier, auditable for aggregate compliance without exposing individual transaction data.

Deployment beyond the Sepolia prototype requires integration with Indonesia's OJK Fintech Sandbox for regulatory approval, training for amil institution staff on wallet management and proof verification workflows, and migration from the mock IDRX stablecoin to a regulated rupiah-backed digital currency.

B. Limitations and Future Work

The circuit currently handles nisab and hawl criteria. The architecture accommodates extension to all eight asnaf categories. Zakat disbursement is performed by the amil institution through the ZakatEscrowManager contract without on-chain verifiability of the disbursement to mustahik (recipients). Future work includes issuing private decentralized identifiers (DIDs) to mustahik for on-chain eligibility attestation and making amil disbursements cryptographically verifiable. A dual governance system combining Sharia Council ZK DAO attestation with Community (Donor) DAO voting is also planned. Other directions include recursive proof systems for multi-period eligibility, compliance-friendly ZK protocols for AML/CFT, and batched settlement optimization [7], [17], [23].

VI. Conclusion

We built and deployed a privacy-preserving zakat protocol on Ethereum Sepolia using Noir circuits and UltraHONK proofs. Existing blockchain zakat systems treat institutional accountability and individual privacy as contradictory. This protocol shows they are compatible. The circuit (29 ACIR opcodes) verifies nisab and hawl eligibility client-side. The Pedersen-hash nullifier prevents double-claims without deanonymizing recipients. The four-tier privacy model lets donors choose their disclosure level. Across five benchmark runs, proof generation averaged 246.8 ms (Barretenberg v4.2.1). An end-to-end Sepolia transaction consumed 721,345 gas.

Cryptographic privacy and Sharia compliance are not in tension. Ibn Ashur's framing of *nafs* protection as encompassing financial dignity [6] aligns with ZK proofs: the proof says "this recipient is eligible" without revealing identity, income, or assets. The `ZKDonationReceived` event gives amil institutions an auditable record of aggregate zakat flow, while the nullifier registry prevents abuse. The blockchain's public record and what it reveals about individuals are decoupled, for the first time in a zakat-specific protocol.

Future work extends the circuit to all eight asnaf categories and adds recipient-side encrypted note disbursement. OJK Fintech Sandbox integration for production deployment is also planned.

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AI-assisted tools (OpenAI Codex, Claude Code) were used for code scaffolding and literature review synthesis. AI-generated text has been human-paraphrased throughout. Residual AI patterns comprise approximately 14% of the document. All core algorithmic contributions, protocol design, security analysis, and implementation were developed and verified by the authors.

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